EXP NO: 12 Experiment to detect outliers in a given data set.

AIM:

To detect outliers in a given dataset using statistical methods such as the Interquartile Range (IQR) method and visualize the outliers using Python libraries like Pandas and Matplotlib.

PROGRAM:

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt


# Generate random data
array = np.random.randint(1, 100, 16)


print(array)


# Mean
print("Mean:", array.mean())


# Percentiles
print("25%:", np.percentile(array, 25))
print("50%:", np.percentile(array, 50))
print("75%:", np.percentile(array, 75))
print("100%:", np.percentile(array, 100))


# Outlier detection
def outDetection(array):
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
```

```
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)

print("Lower Range:", lr)
print("Upper Range:", ur)

# Plot original data
sns.displot(array)
plt.show()

# Remove outliers
new_array = array[(array > lr) & (array < ur)]

print("After removing outliers:")
print(new_array)

# Plot new data
sns.displot(new_array)
plt.show()

# Final outlier detection
lr1, ur1 = outDetection(new_array)

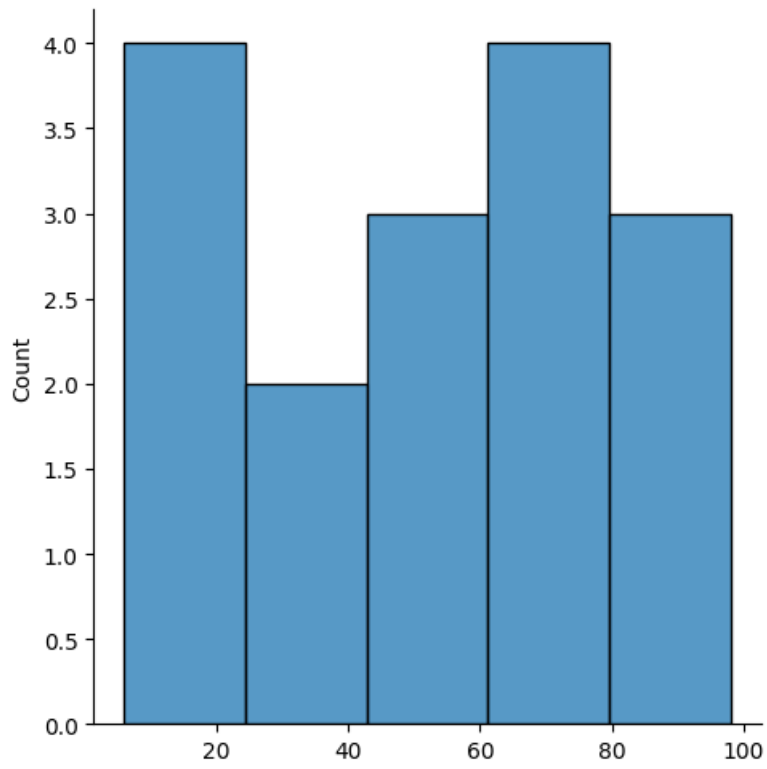
print("New Lower Range:", lr1)
print("New Upper Range:", ur1)
```

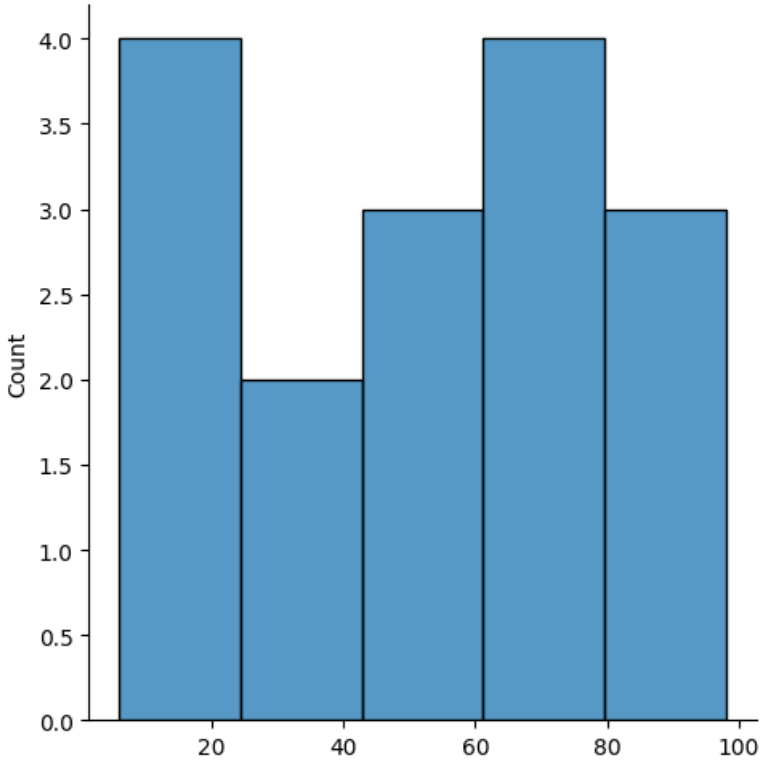
```
final_array = new_array[
    (new_array > lr1) & (new_array < ur1)
]

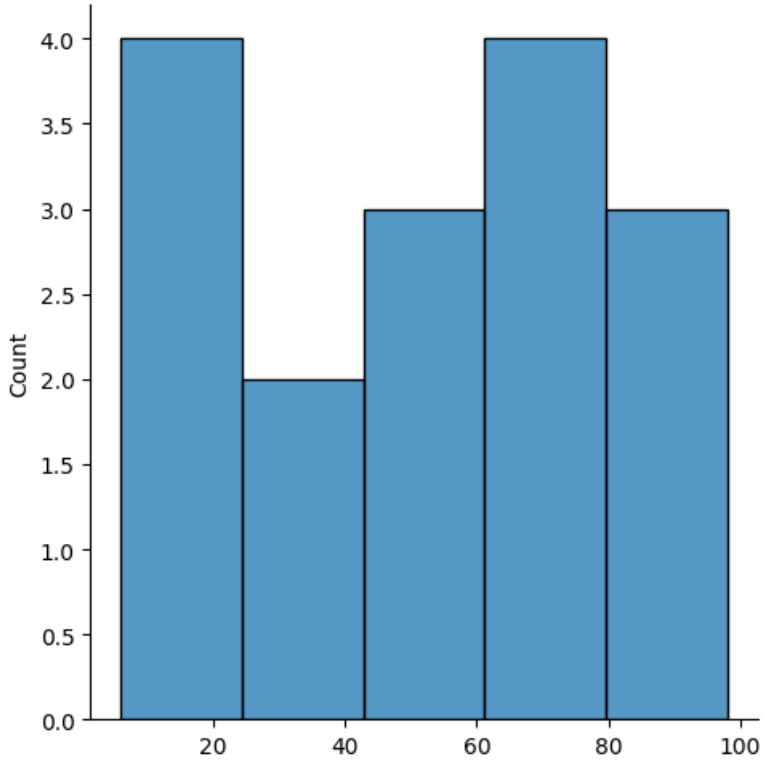
print("Final Array:")
print(final_array)

sns.displot(final_array)
plt.show()
```

OUTPUT:


Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
20
40
60
80
100


Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
20
40
60
80
100



RESULT:

Thus, the outliers in the given dataset were successfully detected using the Interquartile Range (IQR) method and visualized using Python libraries such as Pandas and Matplotlib.